UNITED STATES
NUCLEAR REGULATORY COMMISSION
WASHINGTON, D.C. 20555-0001

March 14, 2019

Mr. Scott Borland
Quality Assurance Manager
HydroAire Services, Inc.
834 W. Madison
Chicago, IL 60607

SUBJECT: NUCLEAR REGULATORY COMMISSION VENDOR INSPECTION REPORT OF HYDROAIRE SERVICES, INC., NO. 99902072/2019-201, NOTICE OF NONCONFORMANCE

Dear Mr. Borland:

From January 28 through February 1, 2019, the U.S. Nuclear Regulatory Commission (NRC) staff conducted an inspection at HydroAire Services, Inc. (hereafter referred to as HydroAire) facilities in Chicago, IL. The purpose of this limited-scope routine inspection was to assess HydroAire's compliance with the provisions of Title 10 of the *Code of Federal Regulations* (10 CFR) Part 21, "Reporting of Defects and Noncompliance," and selected portions of Appendix B, "Quality Assurance Program Criteria for Nuclear Power Plants and Fuel Reprocessing Plants," to 10 CFR Part 50, "Domestic Licensing of Production and Utilization Facilities."

This technically-focused inspection specifically evaluated HydroAire's implementation of the quality activities associated with design, fabrication, assembly, and testing of safety-related pumps being supplied to the U.S. operating nuclear power plants. The enclosed report presents the results of the inspection. This NRC inspection report does not constitute NRC endorsement of HydroAire's overall quality assurance (QA) or 10 CFR Part 21 programs.

Based on the results of this inspection, the NRC inspection team found that the implementation of your QA program did not meet certain regulatory requirements imposed on you by your customers or NRC licensees. Specifically, the NRC inspection team determined that HydroAire was not fully implementing its QA program in the areas of control of measuring and test equipment and instructions, procedures, and drawings. The specific finding and references to the pertinent requirements are identified in the enclosures to this letter. In response to the enclosed notice of nonconformance (NON), HydroAire should document the results of the extent of condition review for the finding and determine if there are any effects on other safety-related components. Please provide a written statement or explanation within 30 days of this letter in accordance with the instructions specified in the enclosed NON. We will consider extending the response time if you show good cause for us to do so.

In accordance with 10 CFR 2.390, "Public Inspections, Exemptions, Requests for Withholding," of the NRC's "Rules of Practice," the NRC will make available electronically for public inspection a copy of this letter, its enclosure, and your response through the NRC Public Document Room or from the NRC's Agencywide Documents Access and Management System, which is

accessible at http://www.nrc.gov/reading-rm/adams.html. To the extent possible, your response, should not include any personal privacy, proprietary, or Safeguards Information (SGI) so that it can be made available to the public without redaction. If personal privacy or proprietary information is necessary to provide an acceptable response, please provide a bracketed copy of your response that identifies the information that should be protected and a redacted copy of your response that deletes such information. If you request that such material be withheld from public disclosure, you must specifically identify the portions of your response that you seek to have withheld and provide in detail the bases for your claim (e.g., explain why the disclosure of information would create an unwarranted invasion of personal privacy or provide the information required by 10 CFR 2.390(b) to support a request for withholding confidential commercial or financial information). If SGI is necessary to provide an acceptable response, please provide the level of protection described in 10 CFR 73.21, "Protection of Safeguards Information: Performance Requirements."

Sincerely,

/RA/

Kerri A. Kavanagh, Chief
Quality Assurance Vendor Inspection Branches 1 and 2
Division of Construction Inspection
and Operational Programs
Office of New Reactors

Docket No.: 99902072

EPID No.: I-2019-201-0024

Enclosure:
1. Notice of Nonconformance
2. Inspection Report No. 99902072/2019-201
   and Attachment

NOTICE OF NONCONFORMANCE

HydroAire Service Inc.
834 W. Madison
Chicago, IL 60607

Docket No. 99902072
Report No. 2019-201

Based on the results of a U.S. Nuclear Regulatory Commission (NRC) inspection conducted at the HydroAire Service Inc.'s (hereafter referred to as HydroAire) facility in Chicago, IL from January 28, 2019, through February 1, 2019, it appears that HydroAire did not conduct certain activities in accordance with NRC requirements that were contractually imposed upon HydroAire by its customers or NRC licensees:

A. Criterion XII, "Control of Measuring and Test Equipment," of Appendix B, "Quality Assurance Criteria for Nuclear Power Plants and Fuel Reprocessing Plants," to Title 10 of the *Code of Federal Regulations* (10 CFR) Part 50, "Domestic Licensing of Production and Utilization Facilities," states in part that, "Measures shall be established to assure that tools, gages, instruments, and other measuring and testing devices used in activities affecting quality are properly controlled, calibrated, and adjusted at specified periods to maintain accuracy within necessary limits."

   Criterion V, "Instructions, Procedures, and Drawings," of Appendix B to 10 CFR Part 50, states that "Activities affecting quality shall be prescribed by documented instructions, procedures, or drawings, of a type appropriate to the circumstances and shall be accomplished in accordance with these instructions, procedures, or drawings. Instructions, procedures, or drawings shall include appropriate quantitative or qualitative acceptance criteria for determining that important activities have been satisfactorily accomplished."

   HydroAire's Nuclear Quality Assurance Manual Section 12, "Control of Measuring and Test Equipment," states, in part that, "measuring and test equipment shall be calibrated at intervals established in approved procedures against equipment with higher accuracy levels through an unbroken chain traceable to National Institute of Standard Technology standard or a natural physical constant." The Nuclear Quality Assurance Manual also states that, "pressure gages used for hydrostatic test will be calibrated prior to and after each test or series of tests using the same pressure gage not to exceed two weeks."

   Contrary to the above, as of February 1, 2019, HydroAire failed to establish adequate measures to assure that pressure gages were properly calibrated and adjusted at specific periods to maintain accuracy within necessary limits and failed to establish documented instructions or a procedure that would prescribe the calibration process of pressure gages for use in activities affecting quality.

   Specifically, the NRC inspection team determined that, HydroAire failed to calibrate its digital pressure gages over their entire working range of 0-7000 pounds per square inch gage (psig) used during hydrostatic test activity of Section III, "Rules for Construction of Nuclear Facility Components," of the American Society of Mechanical Engineers (ASME) Boiler & Pressure Vessel (B&PV) Code. The NRC inspectors identified twenty-two hydrostatic tests (two safety-related pumps and eighteen pump components) where HydroAire inadequately calibrated the digital pressure gages at an operating range of 1167-7000 psig and used them during testing of ASME Section III safety-related

components at a hydrostatic pressure range of 152-370 psig. HydroAire has been inadequately calibrating and using these digital pressure gages since 2016. The validity of such hydrostatic test results are of indeterminate quality.

This issue has been identified as Nonconformance 99902072/2019-201-01

Please provide a written statement or explanation to the U.S. Nuclear Regulatory Commission, ATTN: Document Control Desk, Washington, DC 20555-0001, with a copy to the Chief, Quality Assurance Vendor Inspection Branches 1 and 2, Division of Construction Inspection and Operational Programs, Office of New Reactors, within 30 days of the date of the letter transmitting this Notice of Nonconformance. This reply should be clearly marked as a "Reply to a Notice of Nonconformance" and should include for each noncompliance: (1) the reason for the noncompliance or, if contested, the basis for disputing the noncompliance; (2) the corrective steps that have been taken and the results achieved; (3) the corrective steps that will be taken to avoid further noncompliance; and (4) the date when the corrective action will be completed. Where good cause is shown, the NRC will consider extending the response time.

Because your response will be made available electronically for public inspection in the NRC's Public Document Room or from the NRC's Agencywide Documents Access and Management System, which is accessible from the NRC Web site at http://www.nrc.gov/readingrm/adams.html, to the extent possible, it should not include any personal privacy, proprietary, or Safeguards Information so that the NRC can make it available to the public without redaction. If personal privacy or proprietary information is necessary to provide an acceptable response, then please provide a bracketed copy of your response that identifies the information that should be protected and a redacted copy of your response that deletes such information. If you request that such material be withheld, you must specifically identify the portions of your response that you seek to have withheld and provide in detail the bases for your claim of withholding (e.g., explain why the disclosure of information would create an unwarranted invasion of personal privacy or provide the information required by 10 CFR 2.390(b) to support a request for withholding confidential commercial or financial information). If Safeguards Information is necessary to provide an acceptable response, please provide the level of protection described in 10 CFR 73.21, "Protection of Safeguards Information: Performance Requirements."

Dated this the 14th day of March 2019.

U.S. NUCLEAR REGULATORY COMMISSION OFFICE OF NEW REACTORS
DIVISION OF CONSTRUCTION INSPECTION AND OPERATIONAL PROGRAMS
VENDOR INSPECTION REPORT

Docket No.:	99902072
Report No.:	99902072/2019-201
Vendor:	HydroAire Service Inc. 834 W. Madison Chicago, IL 60607
Vendor Contact:	Mr. Scott Borland Quality Assurance Manager Nuclear Division & Corporate E-mail: sborland@hydroinc.com Phone: (312) 738-3000
Nuclear Industry Activity:	HydroAire Service Inc.'s Madison Facility is an ASME N and NPT certificate holder. HydroAire's scope of supply for the commercial US nuclear industry includes manufacturing, repair, and replacement of safety-related American Society of Mechanical Engineers (ASME) Class 1, 2, & 3 and non-ASME safety-related pumps, including spare/replacement parts, components, and appurtenances; associated engineering; and field services.
Inspection Dates:	January 28 – February 1, 2019
Inspectors:	Jonathan Ortega-Luciano NRO/DCIP/QVIB-2, Team Leader Andrea Keim NRO/DCIP/QVIB-2 Raju Patel NRO/DCIP/QVIB-2 Robert J. Wolfgang NRR/DE/EMIB
Approved by:	Kerri A. Kavanagh, Chief Quality Assurance Vendor Inspection Branches 1 and 2 Division of Construction Inspection and Operational Programs Office of New Reactors

EXECUTIVE SUMMARY

HydroAire Services, Inc.
99902072/2019-201

The U.S. Nuclear Regulatory Commission (NRC) staff conducted a vendor inspection at the HydroAire Service Inc.'s (hereafter referred to as HydroAire) facility in Chicago, IL, to verify that it had implemented an adequate quality assurance (QA) program that complies with the requirements of Appendix B, "Quality Assurance Criteria for Nuclear Power Plants and Fuel Reprocessing Plants," to Title 10 of the *Code of Federal Regulations* (10 CFR) Part 50, "Domestic Licensing of Production and Utilization Facilities" and 10 CFR Part 21, "Reporting of Defects and Noncompliance." Furthermore, the NRC inspection team verified that HydroAire had implemented a program in accordance with Section III, "Rules for Construction of Nuclear Facility Components," of the American Society of Mechanical Engineers and Pressure Vessel Code. This was the first NRC vendor inspection at HydroAire.

This technically-focused inspection specifically evaluated HydroAire's implementation of the quality activities associated with the design and fabrication of safety-related pumps and parts being supplied to US operating reactors

These regulations served as the bases for the NRC inspection:

- Appendix B to 10 CFR Part 50
- 10 CFR Part 21

During the course of this inspection, the NRC inspection team implemented inspection procedure (IP) 43002. "Routine Inspections of Nuclear Vendors," dated January 27, 2017, IP 43004, "Inspection of Commercial-Grade Dedication Programs," dated January 27, 2017 and IP 36100, "Inspection of 10 CFR Part 21 and Programs for Reporting of Defects and Noncompliance," dated February 13, 2012.

With the exception of the Nonconformance described below, the NRC inspection team concluded that HydroAire's QA policies and procedures comply with the applicable requirements of Appendix B to 10 CFR Part 50 and 10 CFR Part 21, and that HydroAire's personnel are implementing these policies and procedures effectively. The results of this inspection are summarized below.

10 CFR Part 21

The NRC inspection team reviewed HydroAire's policies and implementing procedures that govern the implementation of its 10 CFR Part 21 program and determined that there is no specific guidance in HydroAire's procedures as to what types of issues would constitute a deviation that would fall into the category of 10 CFR Part 21 noncompliances, or at what point in the corrective action process an identified issue becomes a deviation subject to 10 CFR Part 21 time requirements for evaluation and reportability. While the NRC inspection team did not identify any items in the corrective action program that would have clearly required an evaluation under 10 CFR Part 21, the lack of guidance in this area was considered a minor issue by the NRC inspection team. HydroAire issued corrective action report No. (CAR) 19-03, dated January 30, 2019, to address this issue. No findings of significance were identified.

Commercial-Grade Dedication and Supplier Oversight

The NRC inspection team reviewed HydroAire's policies and implementing procedures that govern the implementation of its commercial-grade dedication and supplier oversight programs to determine compliance with the requirements of Criterion III, "Design Control," Criterion IV, "Procurement Document Control," and Criterion VII, "Control of Purchased Material, Equipment, and Services," of Appendix B to 10 CFR Part 50. The NRC inspection team identified that HydroAire accepted a Nuclear Industry Assessment Committee (NIAC) commercial-grade survey for a heat treatment service supplier that failed to adequately evaluate the critical characteristic (CCs) identified by HydroAire engineers. The NIAC commercial-grade survey was a limited-scope audit of the supplier's QA program rather than commercial-grade survey specific to the CCs of the item or service being dedicated. Furthermore, HydroAire's commercial-grade surveys of heat treatment services performed for two other commercial suppliers incorrectly used the ISO 17025:2005 alternative as the acceptance method for the calibration of the measurement equipment used during the heat treatment. HydroAire initiated CAR No. 19-04, 19-05 and 19-06, dated January 31, 2019, to address these issues. No findings of significance were identified.

Control of Special Processes

The NRC inspection team reviewed HydroAire's policies, implementing procedure and completed reports associated with welding and nondestructive examination (NDE) of components and determined to meet the requirements of Criterion IX, "Control of Special Processes," of Appendix B to 10 CFR Part 50 as well as Section V, "Nondestructive Examination," and Section IX, "Welding and Brazing Qualification," of the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel (B&PV) Code. The NRC inspection team reviewed HydroAire's certification and qualification records of NDE personnel and welders and confirmed they were qualified in accordance with the requirements of ASME Section IX, and Section III of the ASME B&PV Code. No findings of significance were identified.

Design Control

The NRC inspection team reviewed HydroAire's policies and implementing procedures that govern the design control program to verify compliance with the regulatory requirements of Criterion III, "Design Control," of Appendix B to 10 CFR Part 50 and with the applicable requirements of Subsection ND, "Class 3 Components," of the ASME B&PV Code, various editions and addenda. The NRC inspection team verified that the design and procurement specifications were properly translated into HydroAire's specification sheets, drawings, procedures, data sheets, analyses and engineering calculations. No findings of significance were identified.

Test Control

The NRC inspection team reviewed HydroAire's policies and implementing procedures that govern the test control program to verify compliance with the requirements of Criterion XI, "Test Control," and XII, "Control of Measuring and Test Equipment," of Appendix B to 10 CFR Part 50. The NRC inspection team reviewed a sample of job orders and hydrostatic test reports which showed that applicable customer specifications were: correctly translated into job orders; confirmed the test activity was performed in accordance with HydroAire's test procedure; calibrated measuring and test equipment was used; conducted by qualified test personnel;

results were independently verified by a quality control inspector (QCI); and testing was witnessed by authorized nuclear inspector (ANI).

The NRC inspection team issued Nonconformance 999002071/2019-201-01 in association with HydroAire's failure to implement the regulatory requirements of Criteria XII and V, "Instructions, Procedures, and Drawings." Nonconformance 99902071/2019-201-01 cites HydroAire for failing to establish adequate procedures and controls to ensure that the pressure gages used in hydrostatic testing of safety-related components affecting quality were properly calibrated and adjusted at specific periods to maintain accuracy within necessary limits.

Nonconformance and Corrective Action Programs

The NRC inspection team reviewed HydroAire's policies and implementing procedures that govern the implementation of its nonconformance and corrective action program to determine compliance with the requirements of Criterion XV, "Nonconforming Materials, Parts or Components," and Criterion XVI, "Corrective Action," of Appendix B to 10 CFR Part 50. The NRC inspection team also reviewed a sample of items entered into HydroAire's corrective action program during calendar year 2018. The NRC inspection team identified a minor issue related to the corrective action program. The corrective action process control procedure (PCP 116) did not have measures in place to clearly identify how a condition adverse to quality or significant condition adverse to quality are determined and what actions need to be performed once the determination is completed. HydroAire issued CAR No. 19-02, dated January 30, 2019 to address this issue. No findings of significance were identified.

REPORT DETAILS

1. 10 CFR Part 21 Program

   a. Inspection Scope

   The NRC inspection team reviewed HydroAire's policies and implementing procedures that govern the implementation of its 10 CFR Part 21, "Reporting of Defects and Noncompliance," program to determine compliance with regulatory requirements. In addition, the NRC inspection team evaluated the 10 CFR Part 21 postings and a sample of HydroAire's purchase orders (POs) for compliance with the requirements of 10 CFR 21.21, "Notification of Failure to Comply or Existence of a Defect and its Evaluation," and 10 CFR 21.31, "Procurement Documents." The NRC inspection team evaluated whether HydroAire's corrective action and nonconformances were sufficiently integrated such that issues identified in the corrective action program or nonconformances would be appropriately considered for 10 CFR Part 21 evaluation and reportability.

   The NRC inspection team reviewed HydroAire's procedure to perform a 10 CFR Part 21 evaluation and determined that it addresses the requirements for evaluating deviations and failures to comply. The NRC inspection team reviewed only the procedures because at the time of the inspection HydroAire had not performed any evaluations under 10 CFR Part 21.

   The NRC inspection team discussed the 10 CFR Part 21 program with HydroAire's management and technical staff. The attachment to this inspection report lists the documents reviewed and staff interviewed by the NRC inspection team.

   b. Observations and Findings

   The NRC inspection team identified one minor issue concerning the integration between HydroAire's process control procedure (PCP) 116, "Corrective Action," Revision 5, PCP 115, "Control of Non-Conforming Items," Revision 9, and PCP 041, "10 CFR Part 21, Reportability," Revision 8. Specifically, the NRC inspection team identified that although PCP 116 had a note in the procedure regarding the need to check the 'Part 21 Reportable' box on the corrective action form, there was no specific guidance in the procedure as to what types of issues would constitute a potential 10 CFR Part 21 noncompliance, or at what point in the corrective action process an identified issue becomes a deviation subject to 10 CFR Part 21 time requirements for evaluation and reportability. Upon questioning, HydroAire staff indicated that as far as they could remember, no corrective action issue was ever determined to be potentially reportable under 10 CFR Part 21 and no 10 CFR Part 21 evaluations or notifications have been made by HydroAire.

   While the NRC inspection team did not identify any items in the corrective action program that would have clearly required an evaluation under 10 CFR Part 21, the lack of guidance in this area was considered a minor issue by the NRC inspection team. HydroAire issued corrective action report (CAR) No. 19-03, dated January 30, 2019, to address this issue. No findings of significance were identified.

c. Conclusion

The NRC inspection team concluded that with the exception of the minor issue identified herein, HydroAire is implementing its 10 CFR Part 21 program in accordance with the regulatory requirements of 10 CFR Part 21. Based on the limited sample of documents reviewed, the NRC inspection team also determined that HydroAire is implementing its policies and procedures associated with the 10 CFR Part 21 program. No findings of significance were identified.

2. Design Control

a. Inspection Scope

The NRC inspection team reviewed HydroAire's policies and implementing procedures that govern the design control program to verify compliance with the regulatory requirements of Criterion III, "Design Control," of Appendix B, "Quality Assurance Program Criteria for Nuclear Power Plants and Fuel Reprocessing Plants," to 10 CFR Part 50, "Domestic Licensing of Production and Utilization Facilities," (Appendix B to 10 CFR Part 50) and with the applicable requirements of Subsection ND, "Class 3 Components," of the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel (B&PV) Code, various editions and addenda.

The NRC inspection team verified that the design and procurement specifications were properly translated into HydroAire's specification sheets, drawings, procedures, data sheets, analyses and engineering calculations. The specifications verified by the NRC inspection team included material specifications, applicable ASME B&PV Code design and construction requirements, qualification reports, test requirements, and test reports. The associated documentation reviewed included design reports, pump drawings, and lists of materials of construction. All documents reviewed contained appropriate technical details and met the corresponding design specifications.

The NRC inspection team discussed the design control program with HydroAire's management and technical staff. The attachment to this inspection report lists the documents reviewed and staff interviewed by the NRC inspection team.

b. Observations and Findings

No findings of significance were identified.

c. Conclusion

The NRC inspection team concluded that HydroAire is implementing its design control program in accordance with the regulatory requirements of Criterion III of Appendix B to 10 CFR Part 50 and the ASME B&PV Code. Based on the limited sample of documents reviewed, the NRC inspection also determined that HydroAire is implementing its policies and procedures associated with the design control program. No findings of significance were identified.

3. Commercial-Grade Dedication and Supplier Oversight

a. Inspection Scope

The NRC inspection team reviewed HydroAire's policies and implementing procedures that govern the implementation of its commercial-grade dedication and supplier oversight programs to verify compliance with the requirements of Criterion III, "Design Control," Criterion IV, "Procurement Document Control," and Criterion VII, "Control of Purchased Material, Equipment, and Services," of Appendix B to 10 CFR Part 50.

HydroAire is a member of Nuclear Industry Assessment Committee (NIAC), which consists of companies who supply components and services to the nuclear industry based on a quality assurance (QA) program that meets the requirements of Appendix B to 10 CFR Part 50 and accept 10 CFR Part 21. The NRC inspection team confirmed that once a NIAC audit or commercial-grade survey is received, HydroAire's QA Manager reviews them for completeness and adequacy. These documents are evaluated in accordance with HydroAire's QA program and if found to be acceptable, the QA Manager will document and use these evaluations as the basis for including these suppliers on HydroAire's Approved Suppliers List (ASL).

The NRC inspection team reviewed a sample of commercial-grade dedication plans, checklists, reports, associated POs, and commercial-grade surveys of several vendors on HydroAire's ASL. The NRC inspection team evaluated a sample of technical evaluations and concluded that the technical evaluations in the dedication plans appropriately identify the critical characteristics (CCs) and technical attributes necessary to provide reasonable assurance that the services would perform their intended safety function.

The NRC inspection team reviewed a sample of external audits to verify the implementation of HydroAire's supplier oversight program. The NRC inspection team verified that HydroAire had prepared and approved plans that identify the audit scope and applicable checklist criteria before the initiation of the audit activity. The NRC inspection team confirmed that the audit reports contained objective evidence of the review of the relevant QA criteria of Appendix B to 10 CFR Part 50. For audits that resulted in findings, the NRC inspection team verified that the supplier had established a plan for corrective action and that HydroAire had reviewed and approved the corrective action and verified its satisfactory completion and proper documentation. The NRC inspection team verified that the POs included, as appropriate: the scope of work, right of access to facilities, and extension of contractual requirements to subcontractors. In addition, the NRC inspection team confirmed that the reviewed safety-related POs invoked the requirements of Appendix B to 10 CFR Part 50 and 10 CFR Part 21.

The NRC inspection team also reviewed a sample of training and qualification records of lead auditors and verified that auditing personnel had completed the required training and maintained qualification and certification in accordance with HydroAire's policies and procedures.

The NRC inspection team discussed the commercial-grade dedication and supplier oversight programs with HydroAire's management and technical staff. The attachment to this inspection report lists the documents reviewed and staff interviewed by the NRC inspection team.

b. Observation and Findings

The NRC inspection team found three examples where HydroAire did not conduct commercial-grade surveys in accordance with their policies and procedures. The examples are related to commercial services procured by HydroAire for heat treatment services.

In two of the three commercial-grade surveys performed by HydroAire, the NRC inspection team noted that HydroAire listed 'Control of Measurement and Test Equipment,' as a CC. As part of the commercial-grade survey, HydroAire incorrectly allowed two commercial heat treating suppliers to contract the calibration of measurement and test equipment based on the International Organization for Standardization (ISO) 17025:2005 alternative as the acceptance method.

The ISO 17025:2005 alternative is contained in Nuclear Energy Institute (NEI) document No. 14-05A, "Guidelines for the Use of Accreditation in Lieu of Commercial-Grade Surveys for Procurement of Laboratory Calibration and Test Services," Revision 1. The NRC staff considers the alternative to be an acceptable method for licensees and suppliers of basic components to use this process in lieu of performing commercial-grade surveys for procurement of calibration and testing services performed by domestic and international laboratories accredited by ILAC signatories as part of the commercial-grade dedication process. The evaluation by the NRC staff of the NEI guidance is documented in a safety evaluation report dated February 9, 2015 (Agencywide Documents Access Management System Accession No. ML14322A535). The alternative is acceptable for entities having a QA program that meets Appendix B and meets the guidance as approved in the NRC safety evaluation report.

For the third example, the NRC inspection team noted that HydroAire took credit for a NIAC commercial-grade survey to qualify a commercial heat treatment service. The NRC inspection team found that the NIAC commercial-grade survey did not address the CCs of the service being dedicated. Furthermore, the NIAC survey directed users of the survey to address the CCs in its purchase orders. In addition, the evaluation performed by NIAC on this commercial supplier incorrectly allowed the use of the ISO 17025:2005 alternative as the acceptance method for the calibration of the equipment used during the heat treatment.

While HydroAire's procedures for dedicating the services in question were deficient, the NRC inspection team considered this to be a minor issue because HydroAire was able to provide objective evidence to demonstrate that the certificates of calibration for these instruments were up-to-date and were performed using a national standard. Also, HydroAire's fabrication process had provisions in place that would detect if the heat treatment performed on these items was adequate or not. HydroAire initiated CAR Nos. 19-04, 19-05 and 19-06, dated January 31, 2019, to address these issues. No findings of significance were identified.

c. Conclusion

The NRC inspection team concluded that with the exception of the minor issues identified herein, HydroAire is implementing its commercial-grade dedication and supplier oversight programs in accordance with the regulatory requirements of Criterion III, Criterion IV, and Criterion VII of Appendix B to 10 CFR Part 50. Based on the limited sample of documents reviewed, the NRC inspection team also determined that HydroAire is implementing its policies and procedures associated with the commercial-grade dedication and supplier oversight programs. No findings of significance were identified.

4. Control of Special Processes

a. Inspection Scope

The NRC inspection team reviewed HydroAire's policies and implementing procedures that govern the control of special processes to verify compliance with the regulatory requirements of Criterion IX, "Control of Special Processes," in Appendix B to 10 CFR Part 50 and with the requirements in Subsection NCA, "General Requirements for Division 1 and Division 2," Subsection NB, "Class 1 Components," Subsection NC, "Class 2 Components," and Subsection ND, "Class 3 Components," of Section III, "Rules for Construction of Nuclear Facility Components," Section V, "Nondestructive Examination," and Section IX, "Welding and Brazing Qualification," of the ASME B&PV Code, and personnel are qualified in accordance with the American Society for Nondestructive Testing (ASNT) Recommended Practice SNT-TC-1A, "Personnel Qualification and Certification in Nondestructive Testing."

Welding

The NRC inspection team reviewed a sample of HydroAire's welding procedure specifications (WPSs) to ensure appropriate criteria were specified and procedure qualification records (PQRs) to verify they were qualified in accordance with ASME Section IX and Section III Code requirements.

The NRC inspection team witnessed weld activity on HydroAire Order No. NC-8681 for a Retainer Bearing (Spider). The NRC inspection team noted that the welder was following the critical criteria on the WPS listed on the traveler using designated weld wire released by the QCI using a calibrated welding machine.

The NRC inspection team performed a walk-down of the weld storage area to verify weld materials were controlled to prevent degradation, inadvertent use, or loss of traceability in accordance with HydroAire's approved procedures. The NRC inspection team reviewed weld issuance to ensure the job travelers link the weld material to the welder and heat storage. The NRC inspection team noted that the weld area was kept clean and environmentally protected. The NRC inspection team verified the applicable welding data such as weld material and heat/batch number, WPS, inspection procedures used, and the final inspection results were recorded in accordance with the applicable HydroAire procedures.

The NRC inspection team reviewed a sample of certified material test reports (CMTRs) for the weld filler metal used to verify that the material specifications for physical and

chemical properties met the ASME B&PV Code Section II, “Materials,” and Section III requirements. The NRC inspection team confirmed the weld wire were procured from an approved supplier listed on HydroAire’s approved supplier list.

Nondestructive Examination

Although there were no nondestructive examination (NDE) activities during the week of the inspection, the NRC inspection team reviewed HydroAire’s magnetic particle (MP), Ultrasonic Testing (UT), and Liquid Penetrant (LP) procedures to ensure conformance to ASME B&PV Code Section V requirements. The NRC inspection team selected a sample of LP reports for ASME Section III Code safety-related components and verified the examinations were performed by qualified personnel, using qualified procedures, calibrated equipment, and in compliance with the requirements of Section III and V of the ASME B&PV Code.

Qualification and Training Records of Welders and Nondestructive Testing Personnel

The NRC inspection team reviewed training and qualification records for two welders and confirmed the welders had completed all the required training and had maintained their qualification in accordance with HydroAire’s policies and procedures and were qualified in accordance with the requirements of Sections III and IX of the ASME B&PV Code.

The NRC inspection team reviewed HydroAire’s procedure for certification and qualification of NDE personnel and confirmed the procedure was consistent with the latest revision of the ASNT Recommended Practice SNT-TC-1A and Section III of the ASME B&PV Code. The NRC inspection team reviewed training and qualification records for four NDE Level II and one Level III inspector and confirmed that their records reflect the individuals were trained and qualified in accordance with HydroAire’s procedure. The NRC inspection team also verified that all NDE personnel annual vision and acuity records were current. The NRC inspection team verified that the NDE instrumentation used during visual, MP and LP examination were identified with a calibration sticker and verified its calibration record were current, within the range and frequency maintained. In addition, the NRC inspection team verified that the NDE consumable materials used for the LP process which includes penetrant, developer, and cleaner were from the same manufacturer, and confirmed the material test reports specified the halogen and sulfur contents were below 1% by weight in order to meet the requirements of Section V of the ASME B&PV Code.

The NRC inspection team discussed the special processes program with HydroAire’s management and technical staff. The attachment to this inspection report lists the documents reviewed and staff interviewed by the NRC inspection team.

b. Observation and Findings

No findings of significance were identified.

c. Conclusion

The NRC inspection team concluded that HydroAire is implementing its special processes program in accordance with the regulatory requirements of Criterion IX of Appendix B to 10 CFR Part 50. Based on the limited sample of documents reviewed, the NRC inspection team also determined that HydroAire is implementing its policies and procedures associated with the control of special processes program. No findings of significance were identified.

5. Test Control

a. Inspection Scope

The NRC inspection team reviewed HydroAire's policies and implementing procedures that govern the test control program to verify compliance with the requirements of Criterion XI, "Test Control" and Criterion XII, "Control of Measuring and Test Equipment," of Appendix B to 10 CFR Part 50.

During the week of inspection, there was no test activity that the NRC inspection team could witness. Therefore, the NRC inspection team selected a sample of twenty hydrostatic test reports from ASME B&PV Code Section III and safety-related pumps and components supplied to domestic nuclear customers. The NRC inspection team verified the hydrostatic test reports documented the required qualitative and quantitative acceptance criteria and confirmed the test activity was performed in accordance with HydroAire's test procedure, conducted by a qualified test personnel, calibrated M&TE was used, results were independently verified by a quality control inspector (QCI), and witnessed by an ANI.

The NRC inspection team reviewed in-process and finished travelers and noted that the Authorized Nuclear Inspector (ANI) had reviewed and placed witness hold points. The NRC inspectors confirmed the ANI had witnessed the hydrostatic test hold points and signed off on a sample of five nuclear orders.

The NRC inspection team discussed the test control program with HydroAire's management and technical staff. The attachment to this inspection report lists the documents reviewed and staff interviewed by the NRC inspection team.

b. Observations and Findings

The NRC inspection team reviewed calibration records of M&TE selected from a sample of twenty test reports for ASME Section III and safety-related pumps and components. During this review the NRC inspection team noted that HydroAire had used digital pressure gages ID No. 1313113 and 1208422, both having a range of 0-7000 pounds per square inch gage (psig) during hydrostatic testing of two pumps, and eighteen pump components that were safety-related ASME Section III. Upon review of calibration records for ID No. 1313113 and 1208422, the NRC inspection team noted that HydroAire had failed to properly calibrate these digital pressure gages over their working range of 0-7000 psig. HydroAire calibrated the digital gages, using a standard traceable to the National Institute of Standards and Technology (NIST) over an operating range of 1167- 7000 psig. After the calibration had been performed by HydroAire, these gages were used for hydrostatic testing of safety-related ASME Section III components at a

lower range of 152-370 psig. HydroAire has been calibrating and using these digital pressure gages under this condition since 2016.

HydroAire's Nuclear QA manual Section 12, "Control of Measuring and Test Equipment," states, in part that, "measuring and test equipment shall be calibrated at intervals established in approved procedures against equipment with higher accuracy levels through an unbroken chain traceable to NIST standard or a natural physical constant." The Nuclear QA Manual also states, "pressure gauges used for hydrostatic test will be calibrated prior to and after each test or series of tests using the same pressure gauge not to exceed two weeks."

To verify implementation of Section 12 of HydroAire's Nuclear QA Manual, the NRC inspection team reviewed PCP 014, "Instrument Calibration Procedure," that provides controls for HydroAire's M&TE calibration program. However, the procedure has no detailed calibration guidance with respect to gages. Therefore, the NRC inspection team determined that HydroAire did not have a documented instruction or procedure that prescribes the process to calibrate digital pressure gages to ensure they would be calibrated to their full working range and adjusted at specified periods to maintain accuracy within specified limits. Instead, HydroAire has been using Gauge CalXP Pressure Comparator operator instructions during calibration of digital pressure gages. The lack of an adequate procedure resulted in HydroAire failing to establish controls to ensure that the digital pressure gages used in hydrostatic pressure testing of ASME Section III safety-related components affecting quality were properly calibrated and adjusted at specified periods to maintain their accuracy within necessary limits. The NRC inspection team identified this issue as Nonconformance 99902072/2019-201-01 for HydroAire's failure to establish adequate measures to assure that pressure gages were properly calibrated and adjusted at specific periods to maintain accuracy within necessary limits and failed to establish documented instructions or a procedure that would prescribe the calibration process of pressure gages for use in activities affecting quality. HydroAire initiated CAR No. 19-07, dated January 31, 2019, to address these issues.

c. Conclusion

The NRC inspection team issued Nonconformance 99902072/2019-201-01 in association with HydroAire' failure to implement the regulatory requirements of Criterion XII and Criterion V of Appendix B to 10 CFR Part 50. Nonconformance 99902072/2019-201-01 cites HydroAire for failing to establish adequate procedural controls to ensure that the M&TE used in activities affecting quality are properly calibrated and adjusted at specified periods to maintain accuracy within necessary limits using a documented calibration procedure. Specifically, HydroAire, did not calibrate its digital pressure gages to their full operating range of 0-7000 psig and used them during hydrostatic test of ASME Section III safety-related pumps and components at a lower pressure range of 152-370 psig.

6. Nonconforming Materials, Parts, or Components and Corrective Action

a. Inspection Scope

The NRC inspection team reviewed HydroAire's policies and implementing procedures that govern the nonconformances and corrective action programs to verify compliance

with the requirements of Criterion XV, “Nonconforming Materials, Parts, or Components,” and Criterion XVI, “Corrective Action,” of Appendix B to 10 CFR Part 50.

The NRC inspection team reviewed a sample of nonconformance reports (NCRs) to verify that HydroAire: (1) dispositioned the NCRs in accordance with the applicable procedures, (2) documented an appropriate technical justification for various dispositions, and (3) took adequate corrective action regarding the nonconforming items. For NCRs that were dispositioned use-as-is, the NRC inspection team confirmed the technical justifications were documented to verify the acceptability of the nonconforming item. The NRC inspection team also verified that the NCR process provides a link to the 10 CFR Part 21 program.

The NRC inspection team also reviewed a sample of CARs to ensure that conditions adverse to quality were promptly identified and corrected. The NRC inspection team verified the CARs provided: (1) adequate documentation and description of conditions adverse to quality; (2) an appropriate analysis of the cause of these conditions and the corrective actions taken to prevent recurrence, as applicable; (3) direction for review and approval by the responsible authority; (4) a description of the current status of the corrective actions; and (5) the follow-up actions taken to verify timely and effective implementation of the corrective actions. In addition, the NRC inspection team verified that HydroAire’s corrective action program provides a link to the 10 CFR Part 21 program.

The NRC inspection team discussed the nonconformance and corrective action programs with HydroAire’s management and technical staff. The attachment to this inspection report lists the documents reviewed and staff interviewed by the NRC inspection team.

b. Observations and Findings

The NRC inspection team identified a minor issue concerning the integration between the corrective action and nonconformance program with the 10 CFR Part 21 program. Specifically, the NRC identified that both corrective action request and nonconformance forms have a box that quality assurance uses to address the question, “Part 21 Reportable? Yes or No.” The procedures for these forms do not provide any guidance for this box. This question implies that HydroAire has gone through a Part 21 evaluation process and concluded that the issue identified is not reportable to the NRC. The procedures did not address how corrective action requests and nonconformances are screened to determine if the Part 21 process/evaluation is applicable. The NRC did not identify any items in the corrective action program that would have clearly required an evaluation under 10 CFR Part 21, the lack of guidance in the procedures to determine 10 CFR Part 21 applicability was considered a minor issue. HydroAire issued CAR No. 19-03, dated January 30, 2019, to address this issue.

The NRC inspection team identified a second minor issue related to the corrective action program. The HydroAire quality assurance manual discusses the terms of condition adverse to quality (CAQ) and significant condition adverse to quality (SCAQ) but no guidance is provided. The corrective action procedure (PCP 116) does not have measures in place that provides for HydroAire to clearly identify how a CAQ or

SCAQ are determined and what actions are needed to be performed once the determination is completed. HydroAire issued CAR No. 19-02, dated January 30, 2019 to address this issue.

No findings of Significance were identified

c. Conclusion

The NRC inspection team concluded that with the exception of the minor issues identified herein, HydroAire established its nonconforming materials, parts, or components and corrective action programs in accordance with the regulatory requirements of Criterion XV and Criterion XVI of Appendix B to 10 CFR Part 50. Based on the limited sample of documents reviewed, the NRC inspection team also determined that HydroAire is adequately implementing its policies and procedures associated with the nonconformance and corrective action programs. No findings of significance were identified.

7. Entrance and Exit Meetings

On January 28, 2019, the NRC inspection team discussed the scope of the inspection with Mr. Scott Borland, QA Manager, and other members of HydroAire's management and technical staff. On February 1, 2019, the NRC inspection team presented the inspection results and observations during an exit meeting with Mr. Borland, and other members of HydroAire's management and technical staff. The attachment to this report lists the attendees of the entrance and exit meetings, as well as those individuals whom the NRC inspection team interviewed.